01—input

File: 01—input.kicad_sch

02—notch

File: 02—notch.kicad_sch

03—gain—lowpass

File: 03—gain—lowpass.kicad_sch

04—adc—interface

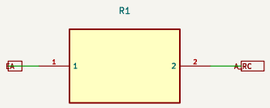
File: 04—adc—interface.kicad_sch

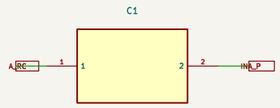
FUNCTIONAL SCHEMATIC DRAFT — NOT PCB READY
Logical pin identifiers; package pins and footprints unassigned.
Read README before editing or PCB export.

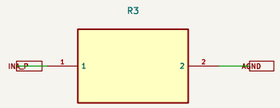
Sheet: /
File: harmonic—detector.kicad_sch

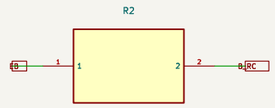
Title:

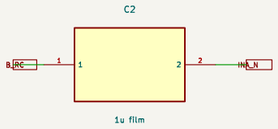
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 5/5

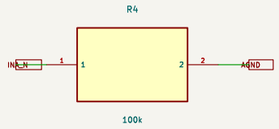


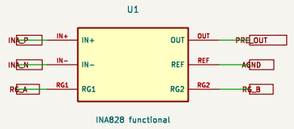


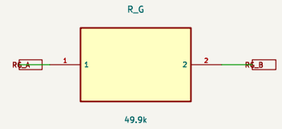












INPUT: EA / EB electrode inputs; DC blocking before amplifier.

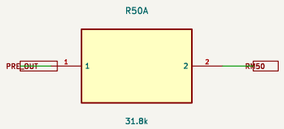
FUNCTIONAL DRAFT: pin numbers are logical names, NOT package pad numbers.

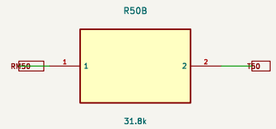
INA828 supply +/−5V and protection require detailed implementation.

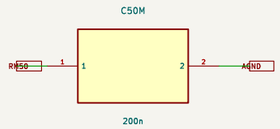
Sheet: /01−input/
File: 01−input.kicad_sch

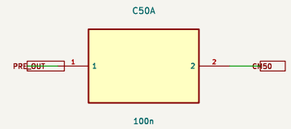
Title:

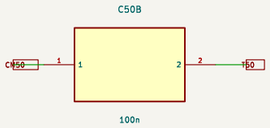
Size: A3 | Date: | **Rev:**
KiCad E.D.A. 9.0.2 | Id: 2/5

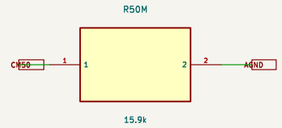


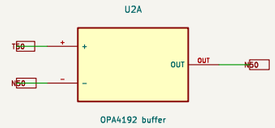


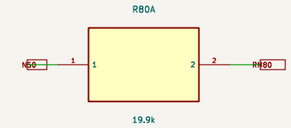


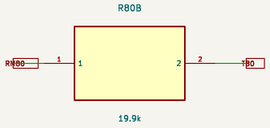


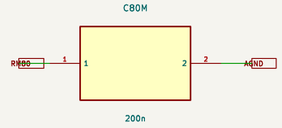


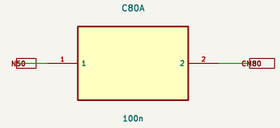


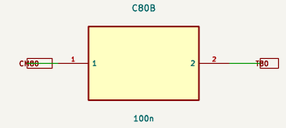


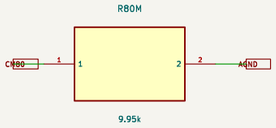


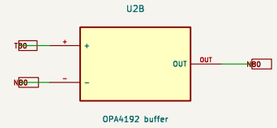










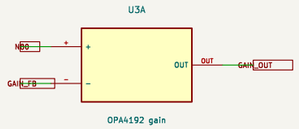


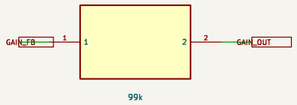
Dual buffered passive twin-T; match R and C ratios and trim.

Passband attenuation at 160/240Hz MUST be calibrated.

Op amp power pins omitted in functional symbols; supply +/-5V.

Sheet: /02-notch/		
File: 02-notch.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 3/5



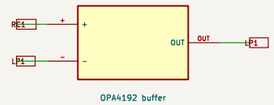




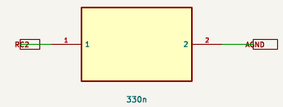


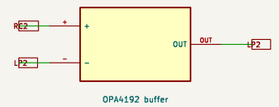




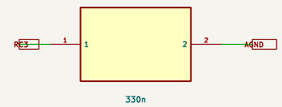


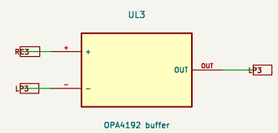


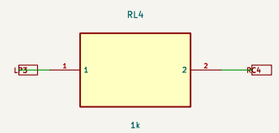


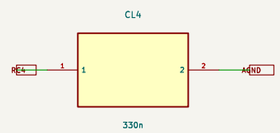


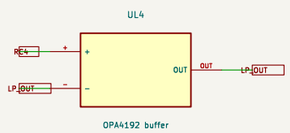










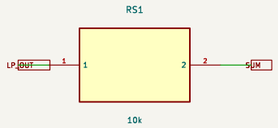


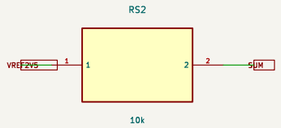
S1 TO ADD: connect GAIN_FB to OPEN / SEL10 / SEL100, mutually exclusive.

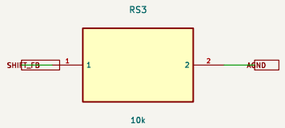
Do not short SEL10 and SEL100 together. Default OPEN gives gain 1.

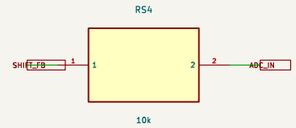
Four independently buffered RC poles, fc=482Hz each.

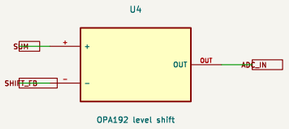
Sheet: /03-gain-lowpass/		
File: 03-gain-lowpass.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.2		Id: 4/5

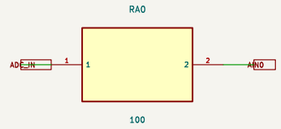


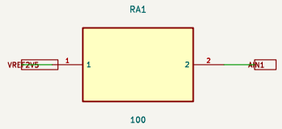


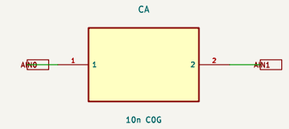












ADS1256 TO ADD: AIN0/AIN1; VREFP=VREF2V5, VREFN=AGND.

AVDD=5V, DVDD=3.3V; REF5025 reference; verify input range and protection.

SPI + DRDY to STM32G4; OLED via I2C; 160/240Hz I/Q detection.

ADC/MCU pin-level circuitry, clocks, power, protection and connectors are NOT implemented.

Sheet: /04-adc-interface/
File: 04-adc-interface.kicad_sch

Title:

Size: A3 | Date: | **Rev:**

KiCad E.D.A. 9.0.2 | Id: 5/5